

SCGNN: Semantic Consistency enhanced Graph Neural Network Guided by Granular-ball Computing

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ABSTRACT

Capturing semantic consistency among nodes is crucial for effective graph representation learning. Existing approaches typically rely on k -nearest neighbors (k NN) or other node-level full search algorithms (FSA) to mine semantic relationships via exhaustive pairwise similarity computation, which suffer from high computational complexity and rigid neighbor selection, limiting scalability and introducing noisy connections. In this paper, we propose the Semantic Consistency enhanced Graph Neural Network (SCGNN), a novel plug-and-play framework that leverages granular-ball computing (GBC) to efficiently capture semantic consistency in a scalable manner. Unlike node-level FSA methods, SCGNN models group-level semantic structure by adaptively partitioning nodes into granular balls, significantly reducing computational cost while improving robustness to noise. To effectively utilize the discovered group-level semantic consistency, we design a dual enhancement strategy. Specifically, (1) a structure enhancement module constructs an anchor-based graph structure, where each anchor is a virtual node representing the group-level semantic carried by a granular ball, then injecting group-level semantic information into the graph structure; and (2) a supervision enhancement module performs label consistency checking (LCC) by combining GBC predictions with model-generated pseudo-labels, thereby producing more reliable supervision signals. SCGNN is compatible with various GNN backbones. During the forward propagation of SCGNN, the vanilla graph and the augment graph are jointly encoded, and their predictions are fused; during the backpropagation, the supervision enhancement module provides enhanced supervision signals to guide parameter updates. Extensive experiments on every benchmarks demonstrate that SCGNN consistently improves performance while maintaining high computational efficiency and scalability.

1. Introduction

Graph-structured data is widely encountered in real-world applications, and graph neural networks (GNNs) Kipf and Welling (2017); Veličković, Cucurull, Casanova, Romero, Liò and Bengio (2018); Gasteiger, Bojchevski and Günnemann (2019) have become a dominant paradigm for learning representations from such data. Depending on the task granularity, GNNs have been successfully applied to node-level tasks Kipf and Welling (2017), edge-level tasks Cai and Ji (2020); Zhang and Chen (2018), and graph-level tasks Xu, Hu, Leskovec and Jegelka (2019); Han, Jiang, Liu and Hu (2022); Zhang, Cui, Neumann and Chen (2018). Despite their differences, these approaches generally rely on learning informative node embeddings through message passing over graph structure.

However, such structure-driven message passing primarily exploits the given graph structure and may fail to fully capture rich semantic relationships embedded in node features. Therefore, relying solely on the observed graph structure is insufficient, and mining semantic information from node features has emerged as an important research direction in graph representation learning Wu, Li and Li

(2024a); Wu, Chen, Du, Huang and Wang (2024b); Pei, Wei, Chang, Lei and Yang (2020); Rubio-Madrigal, Jamadandi and Burkholz (2025); Xie, Tam and Lau (2023); Wang, Yang, Zhang, Xia, Yang and Xu (2026).

The semantic mining direction mainly revolve around two key aspects: (1) the extraction of semantic consistency information, i.e., how to efficiently mine high-quality semantic consistency from features, and (2) the utilization of such information, i.e., how to properly incorporate the extracted semantic consistency into graph neural networks (GNNs) to enhance representation learning.

Node-level full search algorithms (FSA) are widely used for semantic mining Li, Yang, Bai and Ma (2024); Xu, Tian, Fan, Yang, Zhang and Cui (2026); Huang, Shen, Shi and Zhu (2024). These methods operate at a very fine granularity, where the semantic consistency (feature similarity) between every pair of samples is computed. Consequently, in the utilization phase, they naturally modify the graph structure based on pairwise similarity. Specifically, for each node, the $topK$ most similar candidates are selected as its neighbors (commonly referred to as k NN or $topK$), or neighbors are filtered based on a predefined similarity threshold.

In theory, FSA can extract the highest-quality semantic consistency information. However, they suffer from notable limitations in both extraction and utilization. On the extraction side, their quadratic complexity leads to prohibitive computational costs as the number of samples increases Barbero, Vellingker, Saberi, Bronstein and Giovanni (2024); Li

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Table 1

Comparison of time and memory overhead of GBC and k NN on datasets of different scales. We have a total of 92160MB (90GB) of RAM memory.

Datasets(scale)	Time(s)		Memory(MB)	
	GBC	k NN	GBC	k NN
Amazon-ratings(24k)	20	12	227	3995
arxiv2023(46k)	30	33	379	10538
ogbn-arxiv(169k)	106	N/A	597	>92160
ogbn-products(2449k)	1035	N/A	7109	>> 92160

et al. (2024). On the utilization side, k NN or *topK* methods rely on a predefined k , which introduces hyperparameter sensitivity and k -value noise Yang, Yan, Pan, Ye and Fan (2023b); Wang et al. (2026). Similarity threshold based methods provide better adaptivity at the node level, but determining an appropriate threshold is often challenging.

To reduce the computational costs of FSA in semantic consistency mining, we introduce a highly efficient approach—Granular Ball Computing (GBC) Xia, Liu, Ding, Wang, Yu and Luo (2019); Xia, Dai, Wang, Gao and Giem (2022); Xia, Shi, Wang, Xie, Wang and Gao (2025); Cheng, Li, Xia, Wang, Huang and Zhang (2023). GBC can alleviate the FSA scalability bottleneck while also mitigate the challenges of hyperparameter selection and k -value noise in the utilization phase.

Unlike node-level FSA, which performs one-shot global computation, GBC adopts an iterative process that progressively splits granular balls (GBs) until all of them satisfy a predefined stopping criterion Xia et al. (2019). The most significant advantage of GBC lies in its scalability. As shown in Table 1, we compare the overhead of GBC and k NN across datasets of different scales. Even with up to 90 GB of RAM, k NN fails to process the million-scale ogbn-arxiv dataset, whereas GBC can efficiently handle the ten-million-scale ogbn-products dataset with low memory overhead even lower than the overhead of k NN on the much smaller arxiv2023 dataset with only 46k samples. Notably, GBC is slightly slower than k NN only on Amazon-ratings dataset, mainly because k NN benefits from highly optimized matrix operations, whereas GBC incurs additional fixed overhead from its recursive splitting. As the scale increases (e.g., arxiv2023 and beyond), this overhead becomes negligible which makes GBC quickly outperforms k NN.

The substantial gap in computational overhead between k NN and GBC demonstrates the superior scalability of GBC. However, the group-level consistency discovered by GBC cannot be directly used to select neighbors based on pairwise similarity, as in FSA. Therefore, effectively utilizing group-level consistency is the key to integrating GBC into GNNs.

Based on the scalability of GBC and the utilization of its group-level semantic consistency, we propose the Semantic Consistency enhanced Graph Neural Network (SCGNN).

First, we apply GBC to extract group-level semantic consistency. Then, for semantic utilization, we design a dual enhancement strategy consisting of structure enhancement and supervision enhancement.

Specifically, for structure enhancement, we design an anchor-based two-level graph structure construction strategy, where each anchor is a virtual node representing the group-level semantics carried by a granular ball (GB). The two-level graph structure consists of (1) projection edges connecting each anchor to the nodes within its corresponding granular ball, and (2) bridging edges linking anchors that share similar semantics, which inject group-level semantic consistency into the graph structure and enables GNNs to exploit semantic consistency via spectral information. For supervision enhancement, we perform GB-based label prediction, and use the proposed Label Consistency Check (LCC) which combines GB label with the pseudo-label Lee (2013) from model predictions, to produce LCC labels for additional supervision.

In the overall architecture of SCGNN, during the forward propagation, we adopt a parallel prediction and prediction fusion scheme based on both the vanilla graph and the augment graph. During the backpropagation, the model is jointly supervised by train labels and LCC labels. Furthermore, the model is designed in a plug-and-play manner, enabling SCGNN to be compatible with various GNN backbones.

The main contributions are summarized as follows:

1. **A new perspective for semantic consistency mining.** We introduce granular-ball computing (GBC) into graph representation learning, providing a scalable alternative to node-level FSA for capturing semantic consistency.
2. **A dual enhancement strategy for utilizing group-level semantic consistency.** To effectively exploit the group-level semantic consistency discovered by GBC, we design a dual enhancement strategy: (1) a *structure enhancement* module that constructs an anchor-based two-level graph structure to inject semantic consistency into the graph structure, and (2) a *supervision enhancement* module that combines GB predictions with model-generated pseudo-labels via a Label Consistency Check (LCC).
3. **A plug-and-play GNN framework.** SCGNN is designed in a plug-and-play manner which enable it to be compatible with various GNN backbones.
4. **Extensive experimental validation.** Extensive experiments on multiple benchmark datasets show that SCGNN consistently outperforms state-of-the-art methods while maintaining high computational efficiency and scalability.

2. Related Work

2.1. Semantic consistency research of GNN

Several studies attempt to exploit semantic consistency in node features to augment or refine graph structure. For instance, AM-GCNWang, Zhu, Bo, Cui, Shi and Pei (2020b):

Adaptive Multi-channel Graph Convolutional Networks constructs a feature similarity graph by computing k-nearest neighbors in the vanilla feature space, thereby introducing additional edges beyond the vanilla graph structure to capture potential semantic relations between nodes. GeomGCNPei et al. (2020): Geometric Graph Convolutional Networks embeds nodes into a latent geometric space and reorganizes neighborhoods based on geometric proximity, enabling the model to capture semantic relationships that may not align with the vanilla graph connectivity. IDGLChen, Wu and Zaki (2020): Iterative Deep Graph Learning proposes to iteratively learn an optimized graph structure from node features by refining edge weights based on feature similarity, allowing the model to better capture intrinsic semantic relationships and alleviate the limitations of the vanilla graph structure.

2.2. Granular-ball computing

Granular-Ball Computing (GBC) was first introduced as a data representation and learning framework grounded in Granular Computing Xia et al. (2019). Instead of treating individual samples as the basic computational units, GBC represents data using granular balls, each of which corresponds to a locally semantically consistent region. A granular ball summarizes its contained samples through statistical properties such as the center, radius, and label distribution, enabling learning at a higher level of granularity. This paradigm effectively preserves structural information while improving computational efficiency and robustness to noise.

Subsequent works have focused on improving the construction and scalability of granular balls. Xia et al. (2022, 2025) proposed adaptive granular-ball generation methods that iteratively split the data space based on criteria such as purity and data distribution, enabling flexible and efficient granulation for complex data. These methods significantly enhance the stability and adaptability of GBC in practical scenarios.

To further extend GBC to large-scale unsupervised settings, Cheng et al. (2023) introduced a size-constrained granular-ball construction strategy, where the number of samples within each ball is bounded (e.g., $O(\sqrt{n})$), removing the reliance on label information. Based on this representation, they combined GBC with density peaks clustering, performing density estimation and clustering at the granular-ball level to substantially reduce computational cost on large-scale datasets.

3. Methodology

In this section, we first introduce the notations and functions in section 3.1, then we will introduce our method starting from the GBC semantic consistency estimator in section 3.2, finally, the data enhancement strategy and the semantic consistency graph neural network (SCGNN) will be introduced in section 3.3 and 3.4.

Table 2

Notations and descriptions.

Notation	Description
x	a node
$\mathbf{G}$	the vanilla graph
$\mathbf{G}^{\text{aug}}$	the augment graph
$\mathbf{L}$	the label vector
$\mathbf{F}$	the features matrix
$\mathbf{GBs}$	the generated granular balls set
$\mathbf{GB}_k$	the k -th granular ball
$l(\mathbf{GB}/x)$	the label of $\mathbf{GB}$ or node x
$p(\mathbf{GB})$	the purity of a granular ball
$\text{GNN}(\cdot)$	an classic GNN model
$\text{GBC}(\cdot)$	GBC semantic consistency estimator

3.1. Preliminary

In this section, we give all functions and notations we use in the equations of this paper, presented as Table 2.

We give the functions and notations usually used in GNN fields first. The origin graph $\mathbf{G} = (\mathbf{V}, \mathbf{E})$, $\mathbf{V}$ denotes the node set and $\mathbf{E}$ denotes the edge set, and $\mathbf{G}^{\text{aug}} = (\mathbf{V}^{\text{aug}}, \mathbf{E}^{\text{aug}})$ is the augment graph, $\mathbf{G} \subset \mathbf{G}^{\text{aug}}$. All features are represented by a matrix $\mathbf{F} \in \mathbb{R}^{n \times d}$ where $n = |\mathbf{V}|$ is the number of nodes and d is the number of features. Each node has only one label represented as a label vector $\mathbf{L}$.

$\text{GNN}(\cdot)$ denotes any GNN model, the predictions of GNNs is $\mathbf{P} \in \mathbb{R}^{n \times c}$. $\text{GBC}(\cdot)$ is the semantic consistency estimator guided by the granular ball computing, and the predictions of the estimator is $\mathbf{P}^{\text{gbc}} \in \mathbb{R}^{n' \times c}$.

3.2. GBC semantic consistency estimator

In this section, we will introduce the granular-ball computing (GBC) we used in this paper, and next build on the GBC introduce the GBC semantic consistency estimator.

We feed all samples into GBC in a transductive manner, including features of all samples and labels of the training samples, to perform granular-ball computing, before formulating the GBC, we need to introduce an important threshold index in granular-ball computing which is purity and it is computed as below:

$$p(\mathbf{GB}) = \frac{\max_{y_j \in \mathbf{GB}_k} \sum_{x_i \in \mathbf{GB}} \mathbb{I}(l(x_i) = y_j)}{|\mathbf{GB}|}. \quad (1)$$

In particular, if there is no labeled node inside the $\mathbf{GB}$, the $p(\mathbf{GB})$ will be directly set to 1 to terminate the continued division of it.

For intuitive and rigorous expression, we will introducing the semi-supervised GBC we adapted and used via algorithm 1. The k -means algorithm we use is based on the euclidean distance.

We combine supervised granular-ball computing Xia et al. (2019) and unsupervised granular-ball computing Cheng et al. (2023), the advantage of this is that unsupervised method will split the oversized $\mathbf{GB}$ produced by the

Algorithm 1 GBC

Input: Vanilla features matrix $\mathbf{F}$, Train nodes label $\mathbf{L}^{\text{train}}$.

Parameter: the purity threshold $t = 0.8$.

Output: GBs.

Initialization an queue with one element $\mathbf{GBs} = \{f_1, f_2, \dots, f_n\}$, $f \in \mathbf{F}$.

Do

Let num equal to the length of the queue $\mathbf{GBs}$, i.e., $num = |\mathbf{GBs}|$.

For each $\mathbf{GB}$ **in** $\mathbf{GBs}$

Let $p = p(\mathbf{GB})$, by Eq. 1.

IF $p \leq t$

Let k equal to the classes number of all labeled nodes in current $\mathbf{GB}$.

Dequeue $\mathbf{GB}$ from $\mathbf{GBs}$ queue.

Employ k -means algorithm to divide one $\mathbf{GB}$ into k little $\mathbf{sub_GBs}$.

Enqueue $\mathbf{sub_GBs}$ to $\mathbf{GBs}$.

END

END

While The length of the queue no longer changes, i.e., num equal to $|\mathbf{GBs}|$

Do

Let num equal to the length of the queue $\mathbf{GBs}$, i.e., $num = |\mathbf{GBs}|$.

For each $\mathbf{GB}$ **in** $\mathbf{GBs}$

IF $|\mathbf{GB}| \geq \sqrt{n}$

Dequeue $\mathbf{GB}$ from $\mathbf{GBs}$ queue.

Employ 2-means algorithm to divide one $\mathbf{GB}$ into 2 little $\mathbf{sub_GBs}$.

Enqueue $\mathbf{sub_GBs}$ to $\mathbf{GBs}$.

END

END

While The length of the queue no longer changes, i.e., num equal to $|\mathbf{GBs}|$

supervised method to ensure that the size of the $\mathbf{GB}$ remains uniform to a certain extent.

$\mathbf{GB}$ may have the label, it is computed as follow:

$$l(\mathbf{GB}) = \underset{y_j \in \mathbf{GB}_k}{\operatorname{argmax}} \sum_{x_i \in \mathbf{GB}} \mathbb{I}(l(x_i) = y_j). \quad (2)$$

Traditional Granular Ball Computing (GBC) adopts an inductive paradigm, where only training samples are used to construct granular balls. Each ball is explicitly modeled as $\mathbf{GB} = (c, r, l(\mathbf{GB}))$, c is the center of $\mathbf{GB}$, r is the radius of $\mathbf{GB}$, and test samples are subsequently classified based on their nearest ball.

In contrast, our objective is not to build an explicit granular ball classifier, but to exploit the scalability and robustness of GBC for estimating local semantic consistency. Therefore, we adopt a transductive GBC strategy, where both training and unlabeled samples participate in the granular formation process, and the transductive strategy has another strength, it can directly use the coverage characteristics of

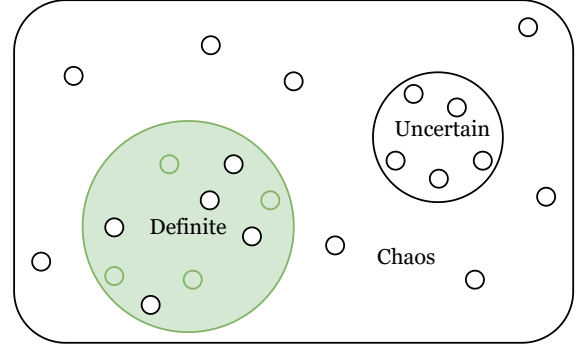


Figure 1: The illustration of three kinds of semantic domain, green circle is definite domain, white circle is uncertain domain and out of circle is chaos domain.

GBC to measure the uncertainty of local semantic consistency. And, owing to this intrinsic properties, all nodes are ultimately assigned to one of three semantic domains: the semantic definite domain: covered by GB; the semantic uncertain domain: covered by GB but without label; and the semantic chaos domain: not covered by any GB; as illustration of Figure 1. This characteristic naturally provides a principled way to quantify the uncertainty of local semantic consistency estimation.

Based on these three semantic domains, we can intuitively and reasonably divide all nodes into two categories: nodes in the semantic definite domain we called it UG(under GB) nodes and the others in the semantic uncertain and semantic chaos domain we called it OOG(out of GB) nodes.

UG nodes are more likely to share consistent local semantics with the labeled samples inside the same ball. In contrast, OOG nodes typically correspond to semantically ambiguous or feature-complex instances that are difficult to separate based solely on feature similarity, these instances tend to exhibit higher noise levels. Due to the inherent robustness of GBC, such noisy or uncertain samples can be effectively identified through the coverage mechanism.

Based on the motivations above, the GBC semantic consistency estimator will be introduced. We provide a formal description of the GBC semantic consistency estimator by integrating the semi-supervised granular-ball computing method introduced above. The GBC (algorithm 1) denoted as $\mathbf{GBC}(\cdot)$ receive the features of samples $\mathbf{F}$ and the supervision information $\mathbf{L}^{\text{train}}$ and last return the $\mathbf{GBs}$ after computing.

$$\mathbf{GBs} = \mathbf{GBC}(\mathbf{F}, \mathbf{L}^{\text{train}}). \quad (3)$$

According the semantic consistency, the label $l(\mathbf{GB})$ i.e., the most common supervised label in each $\mathbf{GB}$ can be treated as the label of all non-label nodes within this $\mathbf{GB}$, simply put, each $\mathbf{GB}$ is treated as a local label domain.

$$\mathbf{p}_j^{\text{gbc}} = l(\mathbf{GB}), j \in \mathbf{GB}. \quad (4)$$

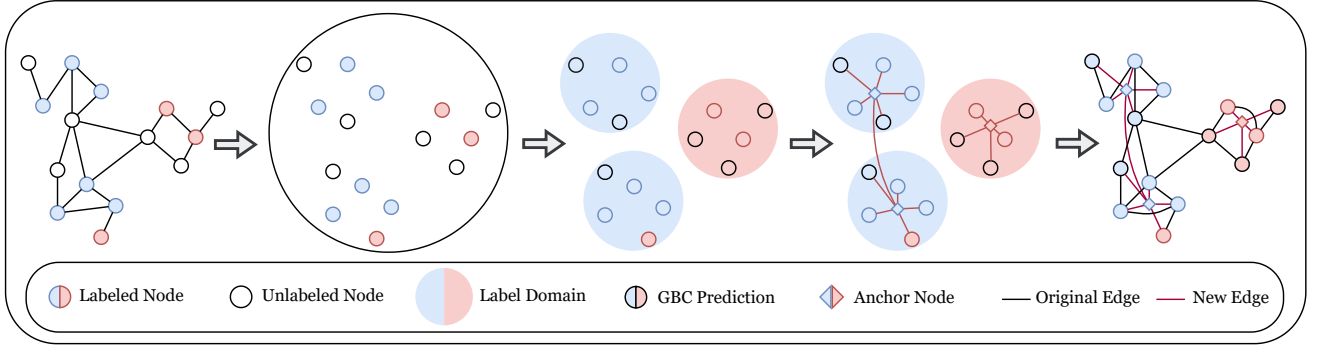


Figure 2: The process of the structure enhancement, four arrows represent GB initialization, GBC predicting, new edges generation and output augment graph.

$\mathbf{j}$ is the sample under **GB**, and $\mathbf{p}_j^{\text{gbc}} \in \mathbb{R}^{n' \times c}$, because of the highly uncertainty of OOG nodes, we gave up label prediction for OOG nodes, so n' is less than the total sample number n , this will make the predictions of GBC much more reliable.

3.3. Semantic consistency enhancement strategy

In this section we will introduce our data enhancement strategy, it contains two parts, one is the structure enhancement and the other is the supervision enhancement, these two will be introduced in turn.

3.3.1. Structure enhancement

The structure enhancement constructs high-quality information propagation pathways based on local semantic consistency, enabling cross-hop connections that capture long-range dependencies beyond the reach of conventional GNNs. The process is illustrated in Figure 2.

Granular-ball computing models feature semantics at the group level. Therefore, explicit pairwise neighbor relations between nodes are not directly available.

So, we design a novel two-level graph structure construction mechanism that converts granular-ball distributions into graph connections. The construction of augment graph consists of two main steps:

First, selecting the UG training nodes belonging to the label of corresponding GB and compute their mean feature as an anchor representing these UG nodes, that is:

$$\mathbf{F}_i^{\text{anchor}} = \frac{1}{n} \sum_{j=1}^n f_j, j \in \mathbf{GB}_i, \mathbf{V}_j \in \mathbf{V}^{\text{train}}. \quad (5)$$

And the label of each anchor is assigned to be the label of corresponding GB:

$$\mathbf{L}_k^{\text{anchor}} = l(\mathbf{GB}). \quad (6)$$

Second, based on the anchors generated above, we construct the high-quality pathways via a two level graph structure, the first level is the projection edges $\mathbf{E}^{\text{project}}$ which is responsible for communicating to the vanilla nodes within

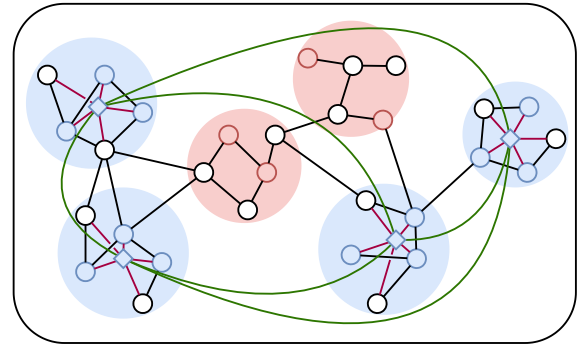


Figure 3: Anchor-based graph structure illustration. For clarity, only the edges associated with blue-labeled nodes is shown. Green edges are bridging edges between anchor nodes, red edges are projection edges.

each **GB** from the anchor, and the second level is the bridging edges $\mathbf{E}^{\text{bridge}}$ which is responsible for connecting all anchors which are belonging to the same label (on large scale dataset, for controlling the number of bridging edges, the full connection could be replaced by random k edges connection).

In this way, vanilla nodes covered by the label domain will get the long distance dependencies in several steps through the high-quality pathways, the illustration is in the Figure 3.

Based on additional anchors and high-quality pathways, the augment graph $\mathbf{G}^{\text{aug}}$ can be constructed as follow:

$$\mathbf{G}^{\text{aug}} = (\mathbf{V}^{\text{aug}}, \mathbf{E}^{\text{aug}}), \quad (7)$$

which $\mathbf{V}^{\text{aug}}$ is the corresponding node samples of the $\mathbf{F}^{\text{aug}}$ and $\mathbf{F}^{\text{aug}} = [\mathbf{F} | \mathbf{F}^{\text{anchor}}]$, $[\cdot | \cdot]$ represent the concatenation. Furthermore, we do not take the radical approach of pruning the vanilla graph structure based on semantic consistency, in other words, the augment graph $\mathbf{G}^{\text{aug}}$ contains the entire vanilla graph $\mathbf{G}$, so the $\mathbf{E}^{\text{aug}} = [\mathbf{E} | \mathbf{E}^{\text{project}} | \mathbf{E}^{\text{bridge}}]$, $[\cdot | \cdot | \cdot]$ represent the concatenation.

3.3.2. Supervision enhancement

The supervision enhancement generates a kind of high-reliability pseudo-label which can provide more supervision information beyond the training label.

Given one GNNs model $\text{GNN}_\theta(\cdot)$, there will be predictions of the input $\mathbf{G}$, i.e.,

$$\mathbf{P} = \text{GNN}_\theta(\mathbf{G}), \mathbf{P} \in \mathbb{R}^{n \times c}, \quad (8)$$

and based on the section of GBC semantic consistency estimator specifically in equation 3 and 4 we got the predictions $\mathbf{P}^{\text{gbc}} \in \mathbb{R}^{n' \times c}$ by the estimator, specially, the number of the predictions of the estimator is n' , less than the predictions of the GNN model due to the uncertainty measure. We finally got two predictions from different paradigms, one is GNN paradigm which usually emphasizes the effect of graph structure, and another is granular-ball computing paradigm which emphasizes the feature semantic and also more robust and explainable. Because these two predictions come from different paradigms, we think they can find different patterns in one same data and then complement each other, so based on $\mathbf{P}$ and $\mathbf{P}^{\text{gbc}}$, we adopt the label consistency checking (LCC) between these two and then get the result $\mathbf{P}^{\text{lcc}}$, that is:

$$\mathbf{P}^{\text{lcc}}_i = \begin{cases} \mathbf{P}_i, & \mathbf{P}_i = \mathbf{P}^{\text{gbc}}_i; \\ \text{N/A}, & \mathbf{P}_i \neq \mathbf{P}^{\text{gbc}}_i. \end{cases} \quad (9)$$

N/A means an invalid prediction that will be filtered, so the result of LCC $\mathbf{P}^{\text{lcc}} \in \mathbb{R}^{n'' \times c}$, the number of $\mathbf{P}^{\text{lcc}}$ will be less than the number of $\mathbf{P}^{\text{gbc}}$, i.e., $n'' \leq n' \leq n$. But the noise of $\mathbf{P}^{\text{lcc}}$ can be lower.

Proposition 1. Consider a c -class classification problem. Assume two predictors produce labels with independent symmetric noise rates R_1 and R_2 . We perform element-wise label consistency checking (LCC) and retain only the samples whose two predicted labels are identical. Let R_3 denote the noise rate among the retained samples. Then

$$R_3 = \frac{\frac{1}{c-1} \cdot R_1 R_2}{(1 - R_1)(1 - R_2) + \frac{1}{c-1} \cdot R_1 R_2}. \quad (10)$$

When $R_1, R_2 \ll 1$, the denominator is dominated by $(1 - R_1)(1 - R_2) \approx 1$, leading to the approximation.

$$R_3 \approx \frac{R_1 R_2}{c - 1}. \quad (11)$$

PROOF OF PROPOSITION1. Let V denote the event that the two predictions agree (i.e., the sample is retained after LCC).

Let N denote the event that the retained label is noisy (i.e., different from the ground-truth label).

By definition, the noise rate among retained samples is

$$R_3 = P(N|V). \quad (12)$$

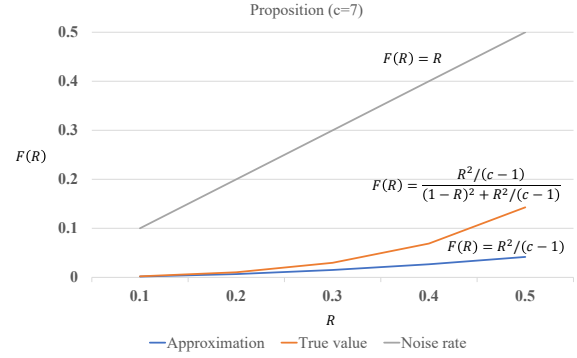


Figure 4: Illustration of the noise reduction effect of LCC with $c = 7$ classes.

Using conditional probability,

$$R_3 = \frac{P(N \cap V)}{P(V)}. \quad (13)$$

A noisy retained label occurs only when both predictors produce the same incorrect label. The probability that both predictors are wrong is $R_1 R_2$. Assuming that incorrect predictions are uniformly distributed over the remaining $c - 1$ classes, the probability that two incorrect predictions coincide is $\frac{1}{c-1}$. Thus,

$$P(N \cap V) = \frac{R_1 R_2}{c - 1}. \quad (14)$$

The retained samples consist of correct agreements and coincident errors:

$$P(V) = (1 - R_1)(1 - R_2) + \frac{R_1 R_2}{c - 1}. \quad (15)$$

Which completes the proof.

DISCUSSION OF PROPOSITION1. This result shows that LCC reduces the noise rate from a linear dependence on individual error rates to a quadratic dependence. Specifically, as shown in Eq. 14, a noisy retained sample occurs only when both predictors make incorrect predictions and their errors coincide, which is significantly less likely than a single prediction error. Therefore, LCC significantly reduces the proportion of noisy labels while preserving correct predictions. Moreover, the reduction effect becomes more pronounced as the number of classes increases, since the coincidence probability is inversely proportional to $c - 1$. This behavior is further validated by the numerical results shown in Figure 4.

This assumption simplifies analysis, although in practice the predictions may exhibit some correlation.

3.4. Semantic consistency enhanced graph neural network

In this section, we rename $\mathbf{P}^{\text{lcc}}$ to $\mathbf{L}^{\text{lcc}}$, because we will treat it as the pseudo label of the samples next.

Based on the augment graph $\mathbf{G}^{\text{aug}}$, anchor label $\mathbf{L}^{\text{anchor}}$, pseudo label $\mathbf{L}^{\text{lcc}}$ and the origin data, the semantic consistency enhancement graph neural network (SCGNN) has been proposed, as illustration in Figure 5.

Starting with forward propagation, SCGNN adopts parallel strategy of vanilla graph $\mathbf{G}$ and augment graph $\mathbf{G}^{\text{aug}}$, and share the parameters of neural network, their corresponding predictions are computed as follow:

$$\begin{cases} \mathbf{P}^{\text{aug}} = \text{GNN}_{\theta}(\mathbf{G}^{\text{aug}}); \\ \mathbf{P} = \text{GNN}_{\theta}(\mathbf{G}), \end{cases} \quad (16)$$

After that, an attention based node-wise weighted sum are utilized for fusing these two kinds of predictions, the fused prediction $\mathbf{P}^{\text{fuse}}$ is computed as:

$$\mathbf{P}^{\text{fuse}} = \text{diag}(\alpha) \cdot \mathbf{P} + \text{diag}(1 - \alpha) \cdot \mathbf{P}^{\text{aug}}, \quad (17)$$

$\text{diag}(\cdot)$ means diagonalization, and $\mathbf{P}^{\text{fuse}}$ is the final prediction of SCGNN, the weights are computed as:

$$\alpha = \frac{\exp(\sigma(\mathbf{P}W_1)W_2)}{\exp(\sigma(\mathbf{P}W_1)W_2) + \exp(\sigma(\mathbf{P}^{\text{aug}}W_1)W_2)}. \quad (18)$$

By the way, the $\mathbf{P}$ used in LCC in equation 9 to generate $\mathbf{L}^{\text{lcc}}$ ($\mathbf{P}^{\text{lcc}}$) comes from equation 16, which is same as the traditional self-train method Lee (2013). And in order to avoid confusion between the two paradigms of GBC and GNN we do not use the $\mathbf{P}^{\text{fuse}}$ to generate the $\mathbf{P}^{\text{lcc}}$.

In backward propagation, there are three losses that supervising the neural network, they are vanilla loss $\text{Loss}_{\text{train}}$, anchor loss $\text{Loss}_{\text{anchor}}$ and the lcc pseudo label loss Loss_{lcc} .

$$\begin{cases} \text{loss}_{\text{lcc}} = \text{CrossEntropy}(\mathbf{P}^{\text{fuse}}, \mathbf{L}^{\text{lcc}}); \\ \text{loss}_{\text{anchor}} = \text{CrossEntropy}(\mathbf{P}^{\text{anchor}}, \mathbf{L}^{\text{anchor}}); \\ \text{loss}_{\text{train}} = \text{CrossEntropy}(\mathbf{P}, \mathbf{L}^{\text{train}}). \end{cases} \quad (19)$$

Specially, the $\mathbf{P}^{\text{anchor}}$ is the subset of $\mathbf{P}^{\text{aug}}$ which is the anchor part exclude the vanilla samples, and the $\mathbf{L}^{\text{lcc}}$ is additional supervisions beyond vanilla train label i.e., $\mathbf{L}^{\text{lcc}} \cap \mathbf{L}^{\text{train}} = \emptyset$.

The total loss is weighted sum by these three parts through the hyper-parameter β and γ , it can be formulated as follow:

$$\text{loss} = \text{loss}_{\text{train}} + \beta \cdot \text{loss}_{\text{anchor}} + \gamma \cdot \text{loss}_{\text{lcc}}. \quad (20)$$

3.5. Complexity analysis

In this section, we analyze the computational complexity of SCGNN from three aspects: (1) time complexity of the preprocessing; (2) time complexity of the model training; (3) space complexity.

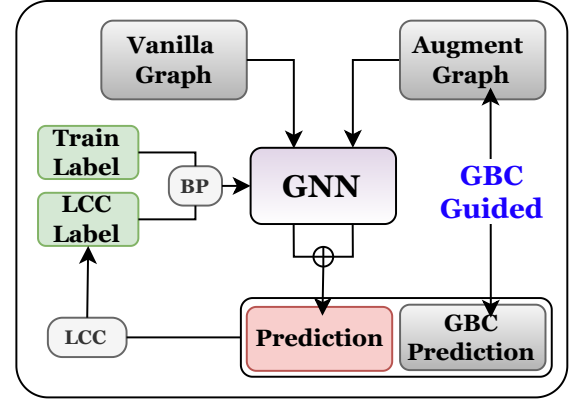


Figure 5: The pipeline of the SCGNN. The augment graph and GBC prediction are guided by granular-ball computing.

3.5.1. Preprocessing complexity

Traditional semantic consistency mining methods, such as k -nearest neighbors (k NN) and other full search algorithms (FSA), rely on exhaustive pairwise similarity computation. Given n nodes with feature dimension d , the similarity matrix construction requires: $\mathcal{O}(n^2d)$. In addition, neighbor selection (e.g., top- k selection) introduces extra sorting or selection overhead, which is typically $\mathcal{O}(nk \log n)$. Therefore, the overall complexity of FSA-based methods is dominated by:

$$\mathcal{O}(n^2d). \quad (21)$$

This quadratic complexity severely limits scalability on large-scale graphs.

In contrast, the proposed GBC adopts an iterative partitioning strategy instead of global pairwise comparison.

Specifically, GBC recursively splits granular balls using local clustering (e.g., k -means). For a granular ball of size m , the cost of one split is: $\mathcal{O}(mdkt)$, where k is the number of clusters and t is the number of iterations.

Since GBC partitions the dataset into multiple granular balls, the total cost can be expressed as: $\sum_i \mathcal{O}(m_i dkt)$, where $\sum_i m_i = n$. Therefore, the overall complexity can be approximated as: $\mathcal{O}(ndkt)$.

In addition, the complexity of anchor-based augment graph mainly consists of three parts:

(1) **Anchor construction.** Each granular ball generates one anchor by averaging node features, which requires a single pass over all nodes: $\mathcal{O}(nd)$; (2) **Projection edges.** Each node is connected to its corresponding anchor, resulting in: $\mathcal{O}(n)$; (3) **Bridging edges.** Anchors belonging to the same class are connected, in the worst case (full connection), the number of edges is: $\mathcal{O}(B^2)$, where B is the number of anchors, and $B = \mathcal{O}(\sqrt{n})$, this yields: $\mathcal{O}(n)$. Overall, the structure enhancement complexity is: $\mathcal{O}(nd)$.

Therefore, the overall complexity of GBC method (include the structure enhancement) is:

$$\mathcal{O}(ndkt), \quad (22)$$

which is near-linear with respect to the number of nodes.

Comparison. Compared with FSA-based methods (Eq. 21), GBC reduces the preprocessing complexity from quadratic to near-linear (Eq. 22), providing a significantly more scalable solution for semantic consistency extraction.

3.5.2. Training complexity

During training, SCGNN performs parallel forward propagation on both the vanilla graph $\mathbf{G}$ and the augment graph $\mathbf{G}_{\text{aug}}$.

For standard message-passing GNNs, the computational complexity is: $\mathcal{O}(|E|d)$. For the augment graph, the cost of one forward pass on $\mathbf{G}_{\text{aug}}$ is: $\mathcal{O}(|E| + n)d$. Since SCGNN processes both graphs, the total training complexity is: $\mathcal{O}(2|E|d + nd)$, which can be simplified to:

$$\mathcal{O}(|E|d). \quad (23)$$

Additional components, such as label consistency checking (LCC) and prediction fusion, introduce only linear overhead $\mathcal{O}(n)$ or $\mathcal{O}(nd)$, which are negligible compared to GNN computation.

3.5.3. Space complexity.

In addition to time complexity, SCGNN also exhibits favorable space complexity. Unlike FSA-based methods that require $\mathcal{O}(n^2)$ memory to store pairwise similarity matrices, GBC operates without constructing such dense structures and only requires $\mathcal{O}(nd)$ memory. This further explains the superior memory efficiency observed in Table 1.

The anchor-based augment graph introduces only $\mathcal{O}(n)$ additional nodes and edges, resulting in an overall space complexity comparable to standard GNNs.

4. Experiments

4.1. Setup

In this section, we will introduce some basic setup of next experiments, train settings, dataset, backbone and baseline will be introduced in turn.

Setting. In all experiments, we chose the adam Kingma and Ba (2014) as optimizer, fixed the max train epochs to 200, fixed the hidden neuro number of backbone to 64, learn rate fixed into 0.01, weight decay fixed into 5×10^{-4} , dropout fixed into 0.5, and in reasoning stage we chose the best performing epoch on the validation set. All experiments were conducted on a machine with 90 GB RAM and an NVIDIA A100 GPU with 40 GB memory.

Dataset. We take 11 datasets in total, including: Cora, Citeseer, Pubmed Yang, Cohen and Salakhudinov (2016), Wiki Yang, Shi, Xiao, Yang, Bhowmick and Liu (2023a), Physics Shchur, Mumme, Bojchevski and Günnemann (2019), Cornell, Wisconsin, Texas Pei et al. (2020), ogbn-arxiv Mikolov, Sutskever, Chen, Corrado and Dean (2013); Wang, Shen, Huang, Wu, Dong and Kanakia (2020a), Roman-empire and Amazon-ratings Platonov, Kuznedelev, Diskin, Babenko and Prokhorenkova (2024). Cora, CiteSeer, PubMed, Cornell, Wisconsin, Texas, Roman-empire, Amazon-ratings and ogbn-arxiv follow the default train splits, others without

default splits we adopt 20% for train, 40% for validate and 40% for test. The more detailed dataset descriptions are organized in Table 3.

Backbone. We take many classic GNN models as the backbone of our SCGNN to validate the efficiency, the GNNs includes: GCN Kipf and Welling (2017), MLP Rumelhart, Hinton and Williams (1986); Werbos (2002), GAT Veličković et al. (2018), APPNP Gasteiger et al. (2019), GIN Xu et al. (2019), GraphSAGE Hamilton, Ying and Leskovec (2017) and H_2GCN Zhu, Yan, Zhao, Heimann, Akoglu and Koutra (2020).

Baseline. All evaluations may be concern with three baseline, and all baselines compared with are all plug-in model as same as ours, including: BORF Nguyen, Hieu, Nguyen, Ho, Osher and Nguyen (2023), ComFy Rubio-Madrigal et al. (2025) and AGMixup Lu, Guan, Zhao, Yang, Zhan, Lu and Tao (2025). All of these baseline methods focus on data augmentation. Specifically, BORF Nguyen et al. (2023) introduced a Batch Ollivier-Ricci Flow-based rewiring algorithm that simultaneously mitigates both over-smoothing and over-squashing issues, thereby enhancing the expressive capability of GNNs. ComFy Rubio-Madrigal et al. (2025) proposed a hybrid rewiring strategy that leverages both community structure and feature similarity, aiming to strengthen local feature coherence while preserving the global community structure to improve GNN performance. AGMixup Lu et al. (2025) presented an adaptive graph mixup framework that employs data mixing techniques to augment graph representations and further boost the generalization ability of GNNs.

4.2. Accuracy evaluation

First of all, we give the numerical evaluations about the accuracy of semi-supervised node classification task, concerned with 8 datasets, the results are taken as average of 10 runs. The numerical statistics can be seen in Table 4. Compared with the SOTA models, our model achieves consistent superiority in accuracy.

4.3. Scalability on large scale datasets

To evaluate the scalability of SCGNN, we conduct experiments on several large-scale datasets, including Amazon-ratings, Physics and ogbn-arxiv. The results are reported in Table 5.

As shown in Table 5, SCGNN consistently improves the performance of the GCN backbone across all datasets where results are available, demonstrating its effectiveness on large-scale graphs.

4.4. Ablation analysis

To gain a deeper understanding of the contributions of different parts of the SCGNN, the ablation analysis will be focus on the structure enhancement, the supervision enhancement and the parallel strategy. The **Abla.1** is W/O the supervision enhancement (LCC labels); the **Abla.2** is W/O the structure enhancement (augment graph); the **Abla.3** is W/O the parallel strategy (vanilla graph channel). Results are shown in Table 6.

Table 3

The details of datasets used in this paper.

Dataset	Nodes	Edges	Features	Class	label rate
Cora	2708	10556	1433	7	5%
CiteSeer	3327	9104	3703	6	4%
PubMed	19717	88648	500	3	0.3%
Wiki	2405	17981	4973	17	20%
Physics	34493	495924	8415	5	20%
ogbn-arxiv	169343	1166243	128	40	54%
Cornell	183	298	1703	5	48%
Wisconsin	251	515	1703	5	48%
Texas	183	325	1703	5	48%
Roman-empire	22662	65854	300	18	50%
Amazon-ratings	24492	186100	300	5	50%

Table 4

The semi-supervised classification accuracy (%) of SCGNN and other all baselines with all seven GNN backbones across all datasets. **Frame** is the overall best across all settings, **bold** is the best in same backbone, underline is the second-best in same backbone, and variance is written at the right bottom corner.

Backbone	Method	Cora	CiteSeer	PubMed	Wiki	Cornell	Wisconsin	Texas	Roman-empire
MLP	BORF	54.9 $\pm$ 0.82	52.31 $\pm$ 0.44	70.38 $\pm$ 0.25	78.48 $\pm$ 0.35	76.21 $\pm$ 1.30	80.39 $\pm$ 0.37	78.38 $\pm$ 0.00	64.57 $\pm$ 0.14
	ComFy	57.20 $\pm$ 0.55	58.96 $\pm$ 0.65	72.64 $\pm$ 0.20	78.86 $\pm$ 0.37	77.84 $\pm$ 4.16	81.18 $\pm$ 1.40	80.00 $\pm$ 1.93	66.03 $\pm$ 0.16
	AGMixup	56.67 $\pm$ 0.17	53.03 $\pm$ 0.66	70.97 $\pm$ 0.90	73.63 $\pm$ 0.91	73.87 $\pm$ 4.59	79.08 $\pm$ 0.92	74.77 $\pm$ 2.55	65.59 $\pm$ 0.05
	SCGNN	67.11 $\pm$ 4.69	63.65 $\pm$ 4.51	75.66 $\pm$ 0.45	79.11 $\pm$ 0.41	78.38 $\pm$ 3.60	82.35 $\pm$ 4.00	83.78 $\pm$ 4.09	66.33 $\pm$ 0.00
GCN	BORF	79.07 $\pm$ 0.20	65.2 $\pm$ 0.42	76.89 $\pm$ 0.15	73.76 $\pm$ 0.36	38.91 $\pm$ 1.6	43.92 $\pm$ 1.7	39.46 $\pm$ 1.4	46.04 $\pm$ 0.46
	ComFy	80.16 $\pm$ 0.66	69.42 $\pm$ 0.51	78.06 $\pm$ 0.39	72.87 $\pm$ 0.53	48.11 $\pm$ 4.93	59.61 $\pm$ 1.52	62.70 $\pm$ 2.37	49.06 $\pm$ 1.76
	AGMixup	81.42 $\pm$ 0.60	68.26 $\pm$ 1.00	77.80 $\pm$ 0.35	66.44 $\pm$ 1.30	40.54 $\pm$ 0.00	52.55 $\pm$ 2.29	60.00 $\pm$ 4.32	43.96 $\pm$ 0.19
	SCGNN	84.00 $\pm$ 0.21	71.23 $\pm$ 2.02	79.71 $\pm$ 0.12	76.61 $\pm$ 0.17	67.57 $\pm$ 1.93	66.67 $\pm$ 1.77	70.27 $\pm$ 2.92	54.43 $\pm$ 0.11
GAT	BORF	76.54 $\pm$ 0.68	66.46 $\pm$ 0.70	75.77 $\pm$ 0.43	68.45 $\pm$ 0.61	45.95 $\pm$ 2.02	43.92 $\pm$ 2.43	40.54 $\pm$ 3.89	52.12 $\pm$ 0.70
	ComFy	73.97 $\pm$ 1.89	66.46 $\pm$ 1.71	75.60 $\pm$ 1.13	56.67 $\pm$ 5.40	43.78 $\pm$ 3.35	49.41 $\pm$ 2.05	52.97 $\pm$ 3.28	58.38 $\pm$ 0.38
	AGMixup	79.63 $\pm$ 0.91	68.70 $\pm$ 0.24	77.17 $\pm$ 1.10	63.58 $\pm$ 2.82	43.78 $\pm$ 6.71	52.55 $\pm$ 7.27	61.62 $\pm$ 6.92	58.23 $\pm$ 0.78
	SCGNN	82.61 $\pm$ 1.99	70.40 $\pm$ 3.77	78.19 $\pm$ 0.26	71.83 $\pm$ 1.02	64.86 $\pm$ 2.21	64.71 $\pm$ 1.36	70.27 $\pm$ 4.02	65.85 $\pm$ 0.02
GIN	BORF	75.91 $\pm$ 0.87	63.18 $\pm$ 1.16	75.45 $\pm$ 1.46	62.33 $\pm$ 0.30	41.89 $\pm$ 1.15	38.82 $\pm$ 1.45	38.65 $\pm$ 1.34	51.45 $\pm$ 0.25
	ComFy	74.92 $\pm$ 0.75	63.94 $\pm$ 0.93	76.86 $\pm$ 0.63	23.51 $\pm$ 2.39	51.89 $\pm$ 1.81	58.04 $\pm$ 4.52	65.95 $\pm$ 2.47	62.94 $\pm$ 3.52
	AGMixup	73.40 $\pm$ 0.29	60.23 $\pm$ 1.07	75.93 $\pm$ 0.66	23.70 $\pm$ 1.47	48.65 $\pm$ 2.21	54.25 $\pm$ 5.15	63.06 $\pm$ 2.55	58.41 $\pm$ 0.71
	SCGNN	78.48 $\pm$ 3.64	68.15 $\pm$ 1.80	78.91 $\pm$ 0.62	63.2 $\pm$ 8.42	70.27 $\pm$ 7.01	72.55 $\pm$ 1.54	72.97 $\pm$ 2.26	63.54 $\pm$ 8.31
APPNP	BORF	79.57 $\pm$ 0.15	66.27 $\pm$ 0.27	76.17 $\pm$ 0.17	75.81 $\pm$ 0.23	38.65 $\pm$ 0.78	47.84 $\pm$ 1.26	41.08 $\pm$ 1.03	58.56 $\pm$ 0.53
	ComFy	80.00 $\pm$ 0.17	69.62 $\pm$ 0.52	78.04 $\pm$ 0.38	74.01 $\pm$ 0.55	54.05 $\pm$ 3.06	63.92 $\pm$ 1.79	58.38 $\pm$ 1.18	58.95 $\pm$ 0.43
	AGMixup	81.87 $\pm$ 0.62	69.87 $\pm$ 1.14	77.67 $\pm$ 0.52	69.89 $\pm$ 1.99	39.64 $\pm$ 1.27	52.94 $\pm$ 4.24	60.36 $\pm$ 2.55	54.42 $\pm$ 0.05
	SCGNN	83.22 $\pm$ 1.72	71.91 $\pm$ 3.59	79.74 $\pm$ 0.11	76.51 $\pm$ 0.24	67.57 $\pm$ 8.77	68.63 $\pm$ 4.00	70.27 $\pm$ 4.21	59.24 $\pm$ 0.04
SAGE	BORF	77.35 $\pm$ 0.35	65.61 $\pm$ 0.72	73.72 $\pm$ 0.56	77.48 $\pm$ 0.14	69.46 $\pm$ 1.72	71.96 $\pm$ 1.57	72.97 $\pm$ 2.02	74.44 $\pm$ 0.28
	ComFy	78.02 $\pm$ 0.83	67.12 $\pm$ 0.71	76.76 $\pm$ 0.67	76.20 $\pm$ 0.79	75.14 $\pm$ 4.69	77.65 $\pm$ 2.38	81.49 $\pm$ 3.06	76.41 $\pm$ 0.32
	AGMixup	79.03 $\pm$ 0.09	68.13 $\pm$ 0.93	77.37 $\pm$ 0.42	67.81 $\pm$ 4.43	65.77 $\pm$ 5.55	77.12 $\pm$ 1.85	78.38 $\pm$ 2.21	76.29 $\pm$ 0.29
	SCGNN	79.73 $\pm$ 2.53	68.20 $\pm$ 2.69	77.62 $\pm$ 0.42	78.59 $\pm$ 0.28	78.38 $\pm$ 4.67	78.43 $\pm$ 1.13	83.78 $\pm$ 7.60	76.80 $\pm$ 0.18
H ₂ GCN	BORF	76.55 $\pm$ 0.43	63.72 $\pm$ 0.42	77.69 $\pm$ 0.58	70.08 $\pm$ 0.80	67.57 $\pm$ 1.53	69.61 $\pm$ 0.62	77.57 $\pm$ 0.78	76.04 $\pm$ 0.18
	ComFy	76.26 $\pm$ 0.77	66.80 $\pm$ 1.52	75.86 $\pm$ 1.20	48.13 $\pm$ 1.97	61.08 $\pm$ 3.93	67.84 $\pm$ 1.40	77.62 $\pm$ 2.82	76.56 $\pm$ 0.35
	AGMixup	78.07 $\pm$ 0.31	63.23 $\pm$ 0.80	76.40 $\pm$ 0.29	71.28 $\pm$ 0.34	66.67 $\pm$ 1.27	64.05 $\pm$ 4.03	77.48 $\pm$ 3.37	76.51 $\pm$ 0.22
	SCGNN	78.10 $\pm$ 0.29	67.31 $\pm$ 0.68	77.91 $\pm$ 2.17	71.41 $\pm$ 2.27	72.97 $\pm$ 2.92	74.51 $\pm$ 0.62	78.38 $\pm$ 2.13	76.89 $\pm$ 5.68

Removing either the supervision enhancement (Abla.1) or the structure enhancement (Abla.2) leads to consistent performance degradation across all datasets. This highlights

the critical role of the proposed GBC-guided semantic consistency enhancement. Specifically, the structure enhancement injects group-level semantic consistency into the graph structure, while the supervision enhancement (LCC) provides reliable training signals to effectively exploit such

Table 5

Performance comparison between vanilla GCN and GCN enhanced with SCGNN on large-scale datasets.

Dataset	GCN	SCGNN
Amazon-ratings	42.93	44.03
Physics	95.98	96.34
ogbn-arxiv	52.01	53.56

Table 6

These ablation experiments are based on GCN backbone, **bold** is the best and underline is the second best.

Dataset	SCGNN	Abla.1	Abla.2	Abla.3
Cora	84.00	<u>82.81</u>	81.65	82.13
CiteSeer	71.23	<u>70.55</u>	70.12	69.10
PubMed	79.71	<u>79.44</u>	79.40	78.82
Wiki	76.61	<u>76.12</u>	75.05	76.09
Cornell	67.57	<u>54.05</u>	54.05	<u>62.16</u>
Wisconsin	66.67	54.95	54.91	<u>56.86</u>
Texas	70.27	<u>67.57</u>	<u>67.57</u>	70.27
Roman-empire	53.38	<u>46.33</u>	48.99	<u>52.92</u>

semantic information. The noticeable performance drops in both cases indicate that semantic consistency must be both properly incorporated into the graph structure and effectively utilized during training.

In contrast, removing the parallel prediction strategy (Abla.3) results in relatively minor performance degradation, suggesting that this component mainly contributes to model robustness by balancing the vanilla and augment graph structures.

Overall, these results validate that the GBC-guided semantic consistency enhancement is the key factor in improving representation learning, while the parallel architecture further stabilizes the model.

4.5. Label noise evaluation

In this section, we evaluate the effectiveness of the proposed Label Consistency Check (LCC) from two perspectives: (1) the quality of pseudo-labels, measured by the noise rate, and (2) the resulting node classification performance. Specifically, we compare three types of predictions: the model prediction $\mathbf{P}$, the granular-ball-based prediction $\mathbf{P}^{gbc}$, and the LCC-refined prediction $\mathbf{P}^{lcc}$. The noise rate is defined as the ratio of incorrectly predicted labels to the total number of nodes. The results are reported in Table 7.

According to the theoretical analysis in Proposition 1, LCC reduces the noise rate from a linear dependence on individual error rates to a quadratic dependence. As shown in Table 7, the noise rate of $\mathbf{P}^{lcc}$ is consistently lower than that of both $\mathbf{P}$ and $\mathbf{P}^{gbc}$ across all datasets. This demonstrates that LCC enhances pseudo-label quality by enforcing agreement between predictions and discarding inconsistent ones.

Furthermore, using $\mathbf{P}^{lcc}$ as additional supervision consistently leads to the best classification performance. In contrast, directly using $\mathbf{P}$ or $\mathbf{P}^{gbc}$ introduces higher noise, which

Table 7

These validation experiments are based on GCN backbone. The indicators contains node classification accuracy and the label noise rate, and the P^{lcc} means using $\mathbf{P}^{lcc}$ as additional label, the P^{gbc} means using $\mathbf{P}^{gbc}$ as additional label, the P means using $\mathbf{P}$ as additional label. **Blod** is the best.

Dataset	ACC(%)			Noise(%)		
	P^{lcc}	P^{gbc}	P	P^{lcc}	P^{gbc}	P
Cora	84.00	82.25	83.81	14	49	21
CiteSeer	71.23	70.12	70.88	36	56	40
PubMed	79.71	79.21	79.68	18	31	24
Wiki	76.61	74.74	74.74	8	17	28
Cornell	67.57	64.86	40.54	21	26	42
Wisconsin	66.67	58.86	56.82	20	29	47
Texas	70.27	67.57	67.57	27	37	42
Roman-empire	53.38	52.43	48.75	19	32	61

negatively affects model training. These results indicate that, although $\mathbf{P}$ and $\mathbf{P}^{gbc}$ may individually contain considerable noise, their agreement subset provides a much more reliable supervision signal.

Overall, the experimental results strongly validate the property of LCC and confirm that it serves as an effective mechanism for generating high-quality pseudo-labels and improving model performance.

4.6. Parameter sensitivity

Our model involves three hyper parameters they are purity p of $\mathbf{GBC}(\cdot)$ and weighted index of two losses β and γ . The purity p is be fixed in 0.8, so in this section we mainly discuss the hyper parameter β and γ . Results in shown in Figure 6.

5. Conclusion

In this paper, we address semantic consistency mining in GNNs by introducing granular-ball computing (GBC) as a scalable alternative to k NN-based full search. By modeling group-level semantics, GBC reduces computational cost and improves robustness.

We further propose a plug-and-play framework, SCGNN, with a dual enhancement mechanism: structure enhancement via anchor-based graph structure construction and supervision enhancement via label consistency checking (LCC).

Extensive experiments demonstrate that SCGNN achieves superior performance with high efficiency. Ablation studies further verify the effectiveness of each component.

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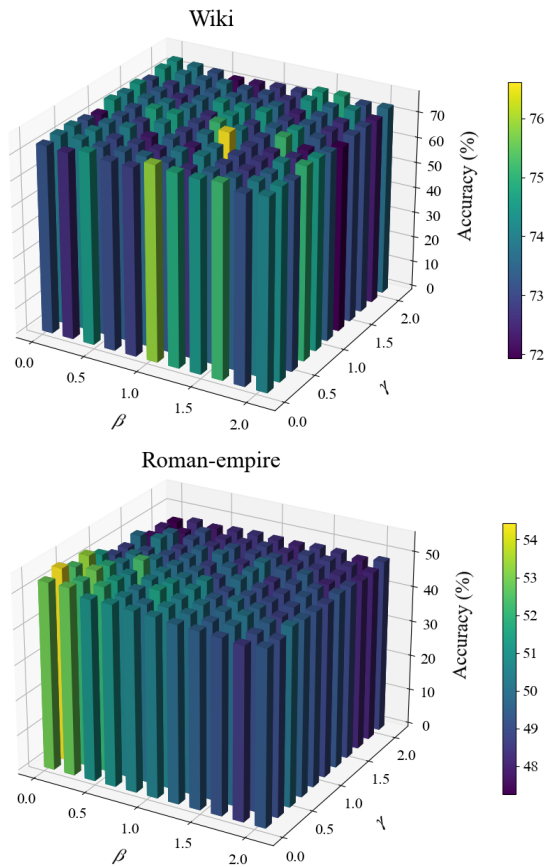


Figure 6: Top Figure is Wiki dataset and the bottom is the Roman-empire dataset.

References

- Barbero, F., Vellingker, A., Saberi, A., Bronstein, M.M., Giovanni, F.D., 2024. Locality-aware graph rewiring in gnns, in: The Twelfth International Conference on Learning Representations, ICLR 2024, Vienna, Austria, May 7-11, 2024, OpenReview.net.
- Cai, L., Ji, S., 2020. A multi-scale approach for graph link prediction, in: Proceedings of the AAAI conference on artificial intelligence, pp. 3308–3315.
- Chen, Y., Wu, L., Zaki, M.J., 2020. Iterative deep graph learning for graph neural networks: Better and robust node embeddings, in: Proceedings of the AAAI Conference on Artificial Intelligence, pp. 3523–3530.
- Cheng, D., Li, Y., Xia, S., Wang, G., Huang, J., Zhang, S., 2023. A fast granular-ball-based density peaks clustering algorithm for large-scale data. *IEEE Transactions on Neural Networks and Learning Systems* 35, 17202–17215.
- Gasteiger, J., Bojchevski, A., Günnemann, S., 2019. Predict then propagate: Graph neural networks meet personalized pagerank, in: International Conference on Learning Representations.
- Hamilton, W., Ying, Z., Leskovec, J., 2017. Inductive representation learning on large graphs.
- Han, X., Jiang, Z., Liu, N., Hu, X., 2022. G-mixup: Graph data augmentation for graph classification, in: International conference on machine learning, PMLR. pp. 8230–8248.
- Huang, J., Shen, J., Shi, X., Zhu, X., 2024. On which nodes does gcen fail? enhancing gcen from the node perspective., in: ICML.
- Kingma, D.P., Ba, J., 2014. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980*.
- Kipf, T.N., Welling, M., 2017. Semi-supervised classification with graph convolutional networks, in: International Conference on Learning Representations.
- Lee, D.H., 2013. Pseudo-label : The simple and efficient semi-supervised learning method for deep neural networks. *ICML 2013 Workshop : Challenges in Representation Learning (WREPL)*.
- Li, L., Yang, W., Bai, S., Ma, Z., 2024. Knn-gnn: A powerful graph neural network enhanced by aggregating k-nearest neighbors in common subspace. *Expert Systems with Applications* 253, 124217. doi:<https://doi.org/10.1016/j.eswa.2024.124217>.
- Lu, W., Guan, Z., Zhao, W., Yang, Y., Zhan, Y., Lu, Y., Tao, D., 2025. Ag-mixup: Adaptive graph mixup for semi-supervised node classification, in: Proceedings of the AAAI Conference on Artificial Intelligence, pp. 19143–19151.
- Mikolov, T., Sutskever, I., Chen, K., Corrado, G.S., Dean, J., 2013. Distributed representations of words and phrases and their compositionality. *Advances in neural information processing systems* 26.
- Nguyen, K., Hieu, N.M., Nguyen, V.D., Ho, N., Osher, S., Nguyen, T.M., 2023. Revisiting over-smoothing and over-squashing using ollivier-ricci curvature, in: Krause, A., Brunskill, E., Cho, K., Engelhardt, B., Sabato, S., Scarlett, J. (Eds.), Proceedings of the 40th International Conference on Machine Learning, PMLR. pp. 25956–25979.
- Pei, H., Wei, B., Chang, K.C.C., Lei, Y., Yang, B., 2020. Geom-gcn: Geometric graph convolutional networks, in: International Conference on Learning Representations.
- Platonov, O., Kuznedelev, D., Diskin, M., Babenko, A., Prokhorenkova, L., 2024. A critical look at the evaluation of gnns under heterophily: Are we really making progress? *arXiv:2302.11640*.
- Rubio-Madriral, C., Jamadandi, A., Burkholz, R., 2025. Gnns getting comfy: Community and feature similarity guided rewiring, in: The Thirteenth International Conference on Learning Representations.
- Rumelhart, D.E., Hinton, G.E., Williams, R.J., 1986. Learning representations by back-propagating errors. *nature* 323, 533–536.
- Shchur, O., Mumme, M., Bojchevski, A., Günnemann, S., 2019. Pitfalls of graph neural network evaluation. *arXiv:1811.05868*.
- Veličković, P., Cucurull, G., Casanova, A., Romero, A., Liò, P., Bengio, Y., 2018. Graph attention networks, in: International Conference on Learning Representations.
- Wang, K., Shen, Z., Huang, C., Wu, C.H., Dong, Y., Kanakia, A., 2020a. Microsoft academic graph: When experts are not enough. *Quantitative Science Studies* 1, 396–413.
- Wang, W., Yang, X., Zhang, Q., Xia, S., Yang, J., Xu, T., 2026. Capturing local and global information: Multi-view graph convolutional network via granular-ball computing and collaborative matrix. *Expert Systems with Applications* 296, 129057. doi:<https://doi.org/10.1016/j.eswa.2025.129057>.
- Wang, X., Zhu, M., Bo, D., Cui, P., Shi, C., Pei, J., 2020b. Am-gcn: Adaptive multi-channel graph convolutional networks, in: Proceedings of the 26th ACM SIGKDD International conference on knowledge discovery & data mining, pp. 1243–1253.
- Werbos, P.J., 2002. Backpropagation through time: what it does and how to do it. *Proceedings of the IEEE* 78, 1550–1560.
- Wu, F., Li, S., Li, S.Z., 2024a. Discovering the Representation Bottleneck of Graph Neural Networks. *IEEE Transactions on Knowledge & Data Engineering* 36, 7998–8008. doi:[10.1109/TKDE.2024.3446584](https://doi.org/10.1109/TKDE.2024.3446584).
- Wu, Z., Chen, Z., Du, S., Huang, S., Wang, S., 2024b. Graph convolutional network with elastic topology. *Pattern Recognition* 151, 110364. doi:<https://doi.org/10.1016/j.patcog.2024.110364>.
- Xia, S., Dai, X., Wang, G., Gao, X., Glem, E., 2022. An efficient and adaptive granular-ball generation method in classification problem. *IEEE Transactions on Neural Networks and Learning Systems* 35, 5319–5331.
- Xia, S., Liu, Y., Ding, X., Wang, G., Yu, H., Luo, Y., 2019. Granular ball computing classifiers for efficient, scalable and robust learning. *Information Sciences* 483, 136–152. doi:<https://doi.org/10.1016/j.ins.2019.01.010>.
- Xia, S., Shi, B., Wang, Y., Xie, J., Wang, G., Gao, X., 2025. Gbct: efficient and adaptive clustering via granular-ball computing for complex data. *IEEE Transactions on Neural Networks and Learning Systems* 36, 12159–12172.

- Xie, S., Tam, D.S.H., Lau, W.C., 2023. Violin: Virtual overbridge linking for enhancing semi-supervised learning on graphs with limited labels, in: Elkind, E. (Ed.), Proceedings of the Thirty-Second International Joint Conference on Artificial Intelligence, IJCAI-23, International Joint Conferences on Artificial Intelligence Organization. pp. 4451–4459. doi:10.24963/ijcai.2023/495. main Track.
- Xu, K., Hu, W., Leskovec, J., Jegelka, S., 2019. How powerful are graph neural networks?, in: International Conference on Learning Representations.
- Xu, T., Tian, G., Fan, J., Yang, X., Zhang, Q., Cui, Y., 2026. Layer embedding deep fusion graph neural network. *arXiv:2604.23324*.
- Yang, R., Shi, J., Xiao, X., Yang, Y., Bhowmick, S.S., Liu, J., 2023a. Pane: scalable and effective attributed network embedding. *The VLDB Journal* 32, 1237–1262. doi:10.1007/s00778-023-00790-4.
- Yang, X., Yan, M., Pan, S., Ye, X., Fan, D., 2023b. Simple and efficient heterogeneous graph neural network, in: Proceedings of the AAAI conference on artificial intelligence, pp. 10816–10824.
- Yang, Z., Cohen, W., Salakhudinov, R., 2016. Revisiting semi-supervised learning with graph embeddings, in: Balcan, M.F., Weinberger, K.Q. (Eds.), Proceedings of The 33rd International Conference on Machine Learning, PMLR, New York, New York, USA. pp. 40–48.
- Zhang, M., Chen, Y., 2018. Link prediction based on graph neural networks. *Advances in neural information processing systems* 31.
- Zhang, M., Cui, Z., Neumann, M., Chen, Y., 2018. An end-to-end deep learning architecture for graph classification, in: Proceedings of the AAAI conference on artificial intelligence.
- Zhu, J., Yan, Y., Zhao, L., Heimann, M., Akoglu, L., Koutra, D., 2020. Beyond homophily in graph neural networks: Current limitations and effective designs.